

AI Software Engineer, Model Training — Jaseci Labs

Ann Arbor / Detroit, MI (hybrid) · Full-time

About the role

We build AI systems that run in production at million-user scale for Fortune 500 companies and fast-growing startups. In this role you will fine-tune and post-train the models those systems depend on, build the agents that use them, and have the opportunity to contribute to and expand Jac, our fast-growing open-source ecosystem and community.

About Jaseci Labs

Jaseci Labs is a team of engineers, professors, and PhDs from the University of Michigan and other universities. We build production AI systems for enterprises and startups, and we created the Jaseci ecosystem and the Jac programming language (jaseci.org), an open-source stack where AI, orchestration, and application logic live in one language. Our production work drives the development of the stack, and the stack lets us deliver faster than a conventional approach.

What you'll do

- **Fine-tune and post-train models.** Prepare training data, run SFT, LoRA, or DPO-style post-training, evaluate the results properly, and deploy the model (quantized, served, and monitored).
- **Ship AI agents to production.** Design, build, and deploy agentic systems, including retrieval, tool use, multi-step workflows, and human-in-the-loop review, and take them all the way to production, where they serve real users and create measurable impact.
- **Work embedded with customer engineering teams.** Learn how large organizations make decisions about AI and what it takes to operate within their constraints.
- **Contribute to the open-source ecosystem.** Help expand Jac and its community: language features, runtime improvements, examples, and documentation that make the stack easier for others to adopt.

What we're looking for

Required

- Hands-on fine-tuning or post-training experience: you have prepared a dataset, trained a model (LoRA, full fine-tune, DPO, or similar), and evaluated it against a baseline
- Hands-on agent-building experience: tool calling, orchestration, retrieval, and evaluation, with any framework or without one

- Strong Python; comfortable with JavaScript or TypeScript
- Experience working with or inside a large organization, or the patience for it
- Degree in Computer Science or a related field, or equivalent experience

Strong plus

- Experience with training and evaluation infrastructure: distributed training, experiment tracking, eval harnesses
- Serving models in production: vLLM, TGI, quantization
- Kubernetes, cloud infrastructure, or MLOps experience
- Experience in regulated industries such as financial services or healthcare
- Open-source contributions of any size
- Familiarity with graph data models or compiler concepts

What we offer

- Real problems at large scale from your first week
- Mentorship from University of Michigan professors and PhDs with production experience
- Ownership of an open-source language and runtime with a growing community
- In-person offices in downtown Ann Arbor and Detroit, with hybrid flexibility
- Competitive compensation and benefits

How to apply

Email community@jaseci.org with the subject line **"AI Software Engineer, Model Training"**. Include a résumé.

Job type: Full-time · Work location: Hybrid, Detroit, MI 48226 and Ann Arbor, MI